1.1.2 Elementary Transformations and Equivalent matrices

Elementary transformations of a matrix

(i) Interchange of any two rows (or columns): $R_i \leftrightarrow R_j \ (or\ C_i \leftrightarrow C_j)$.

(ii) Multiplication of each element of a row (or column) by any non-zero scalar k: $R_i \rightarrow kR_i \ \ (or\ C_i \rightarrow kC_i)$

(iii) Addition to the elements of any row (or column) the same scalar multiples of corresponding elements of any other row (or column):

$$R_i \rightarrow R_i + kR_j . (or\, C_i \rightarrow C_i + kC_j)$$

Equivalent Matrices

Two matrices A and B are said to be equivalent if one is obtained from the another by applying a finite number of elementary transformations and we write it as $A \sim B$ or $B \sim A$.

1.1.3 Echelon form and finding the rank of the matrix (upto the order of 3×4)

A matrix A of order $m \times n$ is said to be in echelon form if

(i) Every row of A which has all its entries 0 occurs below every row which has a non-zero entry.

(ii) The number of zeros before the first non-zero element in a row is less then the number of such zeros in the next row.

Example 1.6

Find the rank of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{pmatrix}$

Solution :

The order of A is 3×3.

$\therefore \rho(A) \leq 3$.

Let us transform the matrix A to an echelon form by using elementary transformations.

Matrix A	Elementary Transformation
$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{pmatrix}$	
$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & -1 & -2 \end{pmatrix}$	$R_2 \rightarrow R_2 - 2R_1$ $R_3 \rightarrow R_3 - 3R_1$
$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$	$R_3 \rightarrow R_3 - R_2$
The above matrix is in echelon form	

The number of non zero rows is 2

$\therefore$ Rank of A *is* 2.

$\rho(A) = 2$.

Note

A row having atleast one non-zero element is called as non-zero row.

Example 1.7

Find the rank of the matrix $A = \begin{pmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{pmatrix}$

Solution:

The order of A is 3×4.

$\therefore \rho(A) \leq 3$.

Let us transform the matrix A to an echelon form

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